CS 6476 PS2

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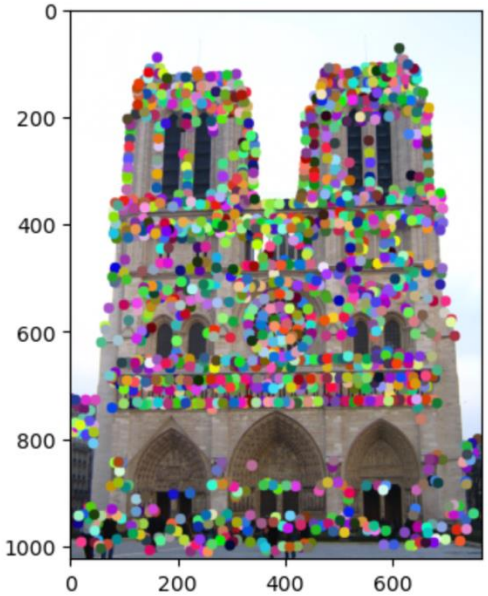
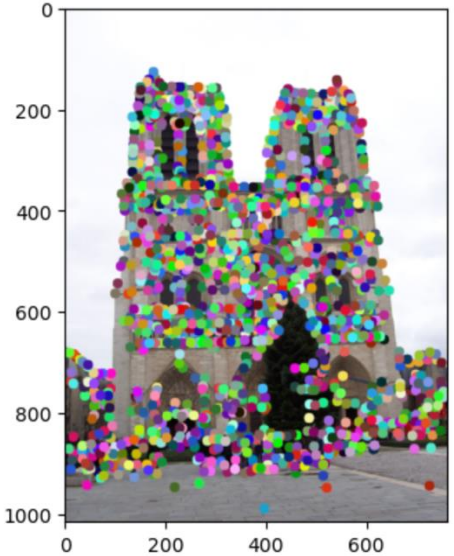
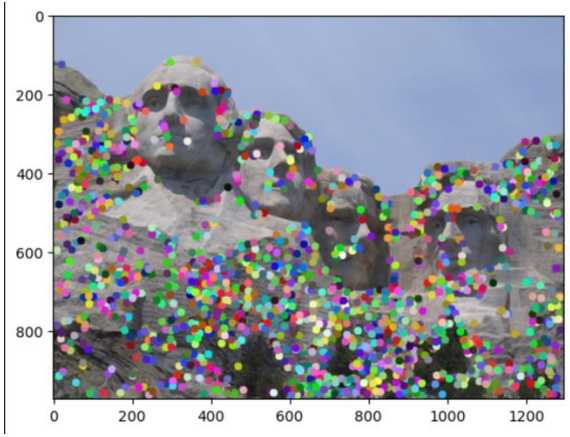
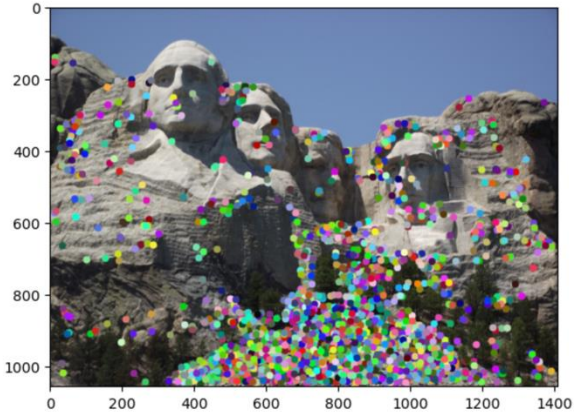
1.1: Harris Corner Detector









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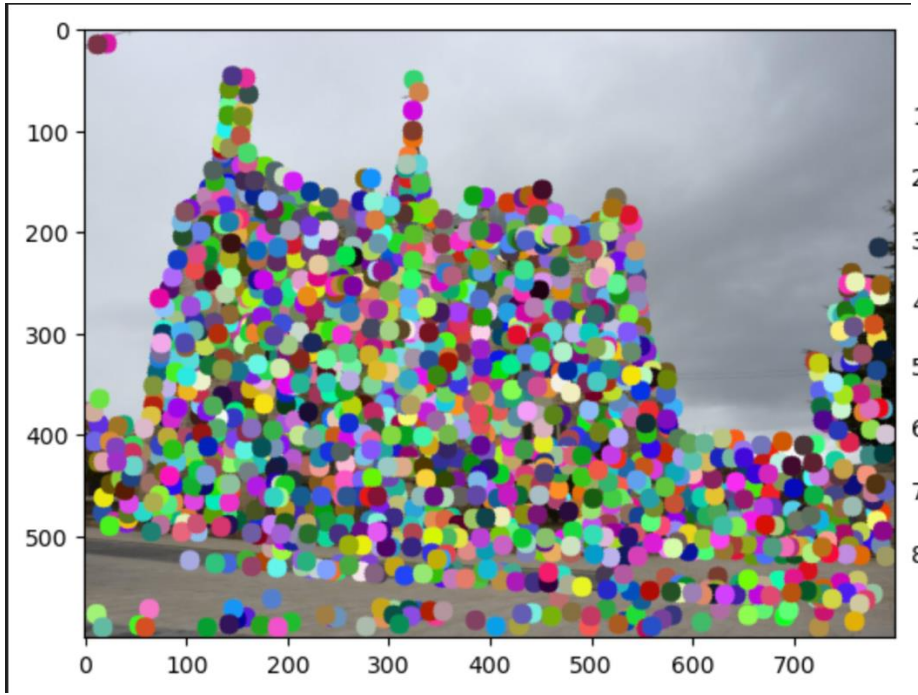
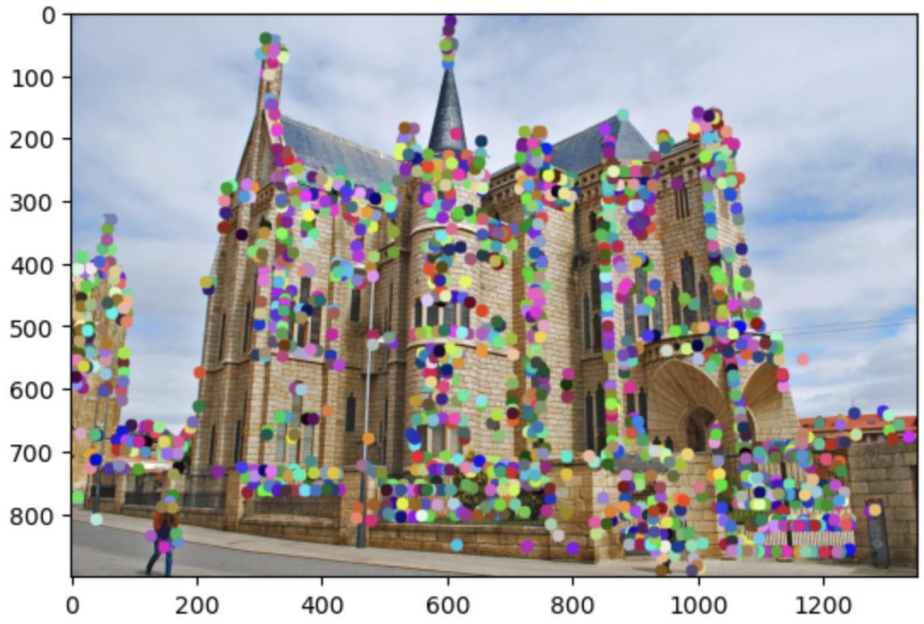




1.1: Harris Corner Detector

- Briefly describe how the Harris corner detector works. [1 pt]

1) Finds points in an image where the intensity changes significantly in two directions. 2) Looks at how the image intensity varies in small neighborhoods 3) summarizes this variation into a simple matrix that captures how strong the changes are 4) scores each pixel based on whether the neighborhood looks like a flat region, an edge, or a corner. Pixels with strong corner scores are kept after filtering out weak or duplicate responses.

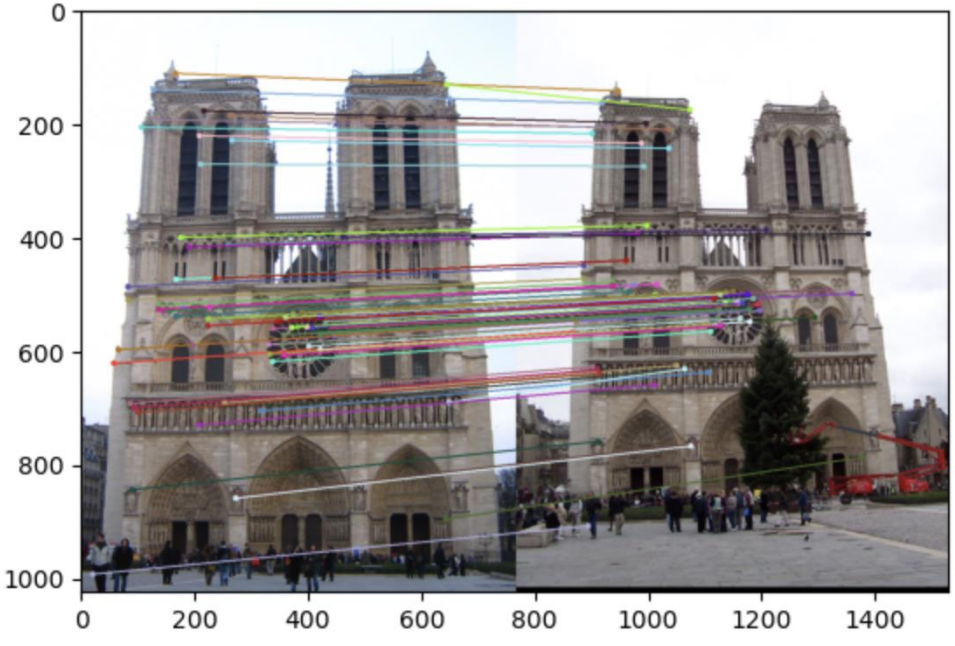
- What does the second_moments() helper function do? [1 pt]

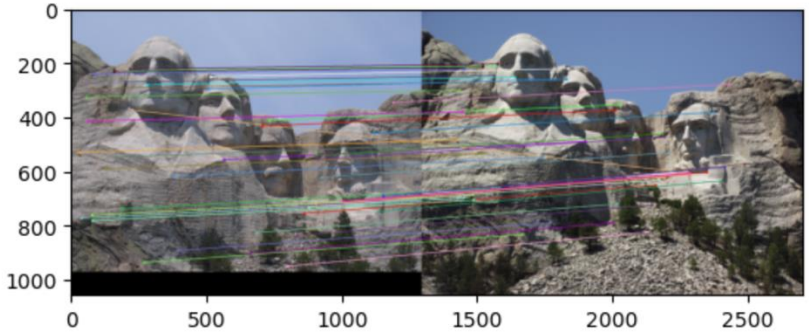
It measures how image intensity changes in a local region by combining information about horizontal and vertical changes in brightness. It smooths these measurements over a small area, so that the result is less sensitive to noise and represents the overall structure around each pixel.

- What does the corner_response() helper function do? [1 pt]

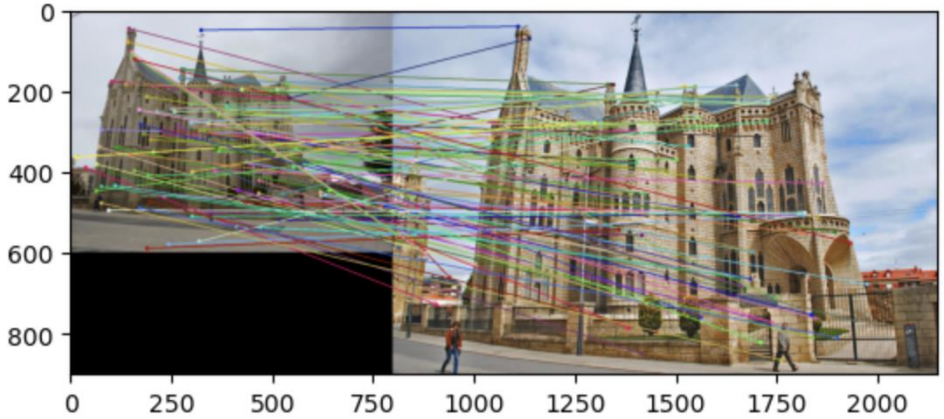
It uses the smoothed gradient information to assign a “corner strength” score to every pixel. A high score means that the local area has strong changes in multiple directions, which is a corner.

1.3: Feature Matching



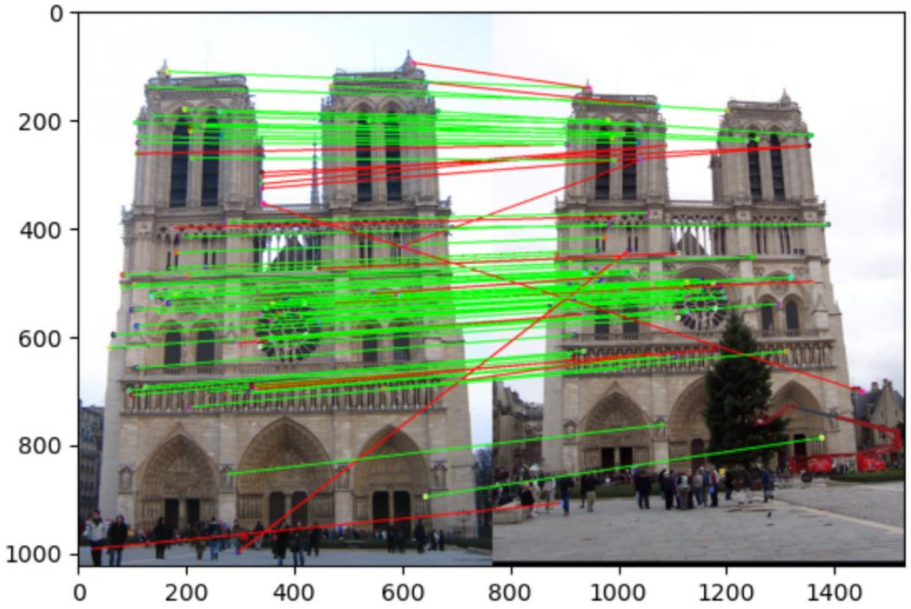


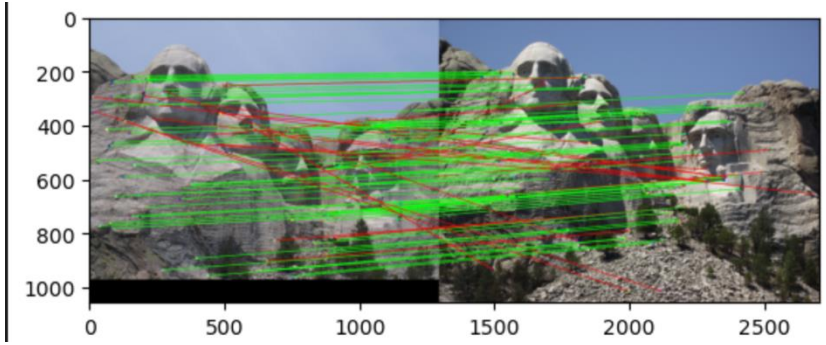
1.3: Feature Matching



First, I compute the Euclidean distance from each descriptor in image 1 to all descriptors in image 2. For each row I take the two smallest distances and apply Lowe's NNDR ratio test, d1/d2 < threshold that I tuned to 0.8. If the test passes, I record a match as [i, j] where i is the index in image 1 and j is the index of the nearest neighbor in image 2. I use the nearest distance d1 as the confidence and sort matches by this value, so the strongest matches come first.

Results: Ground Truth Comparison





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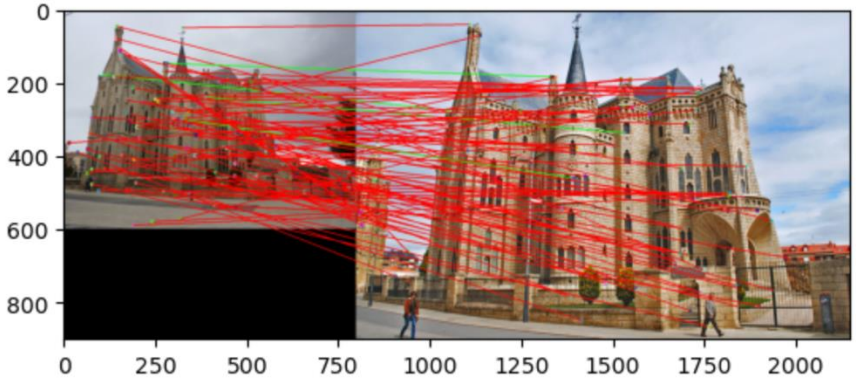

Notre Dame: You found 100/100 required matches
Accuracy = 0.800000
Rushmore: You found 95/100 required matches
Accuracy = 0.750000
Gaudi: You found 100/100 required matches Accuracy = 0.040000

A 2×2 grid is very fast but too coarse, losing spatial detail and giving lower accuracy. The standard 4×4 grid provides a good balance and yields the best performance. Increasing to 5×5 or 7×7 makes descriptors more distinctive but also more sensitive to small misalignments, sometimes improving accuracy on textured scenes but hurting it on repetitive patterns like Gaudi. Very large grids like 15×15 become slow and often overfit noise, with little or no accuracy gain.

1.4(a): Hyperparameter Tuning part 1 [Extra credit]

<Insert images of the ground truth correspondence and their corresponding accuracies for varying sigma in the second moments [3, 6, 10, 30] > [0.5 pts]

When changing the values for large sigma (>20), why are the accuracies generally the same? [0.5 pts]

At large sigma, the gaussian smoothing blur the image so much that fine details are almost gone. The second moment matrix becomes nearly constant in flat areas, so the Harris response mostly depends on big, coarse structures that stay the same across sigma values.

1.4(a): Hyperparameter Tuning part 2 [Extra credit]

<Insert images of the ground truth correspondence and their corresponding accuracies for varying feature width in the SIFT [8, 16, 24, 32] > [0.5 pts]

What is the significance of changing the feature width in SIFT? [0.5 pts]
Feature width controls how much image context each descriptor covers. A small width makes descriptors focus on very local details, while a larger width includes more surrounding pixels. Choosing the right feature width balances distinctiveness and invariance, which directly affects matching accuracy.

1.4(c): Accelerated Matching [Extra credit]

<Insert Runtime/Accuracy of your faster matching implementation. What did you try and why is it faster?> [1 pts]